Remove truncation in hyperbola law for stable matching values in logit markets

ChatGPT (`gpt-5.6-sol`)*

Partial progress generated on 2026-08-24

Abstract

Status: partial progress, not a complete solution of the canonical heterogeneous theorem. The conditional theorem and the homogeneous theorem are complete. The verifier's fundamental objection remains: boundedness and bistochasticity alone have not been shown to imply the required annealed control.

1 The open problem

This paper addresses an open problem posed by Itai Ashlagi, Mark Braverman, Geng Zhao in *Welfare Distribution in Two-sided Random Matching Markets* [1] (Economics and Computation (EC), 2023), Page 7, Section 3.1 (Discussion); reiterated page 18, Section 7.1 (after Proposition 7.3).. The website catalogue entry is problem 4 (`W4383533309_p0`) of the Open Problems in Operations Research collection.

> *Show that the "hyperbola law" for total welfare holds without truncation, namely that*
>
> $$\sup_{\mu \in S} \left| \frac{1}{n} \|X(\mu)\|_1 \|Y(\mu)\|_1 - 1 \right| \xrightarrow{p} 0 \quad \text{as } n \to \infty.$$
>
> *Equivalently, remove the need for δ-truncation in the existing bound $\frac{1}{n}\|X_\delta(\mu)\|_1\|Y_\delta(\mu)\|_1 \approx 1$ that holds uniformly over stable matchings, and prove the corresponding statement for the full sums $\|X(\mu)\|_1$ and $\|Y(\mu)\|_1$.*

The remainder of this document is the partial progress produced by the model, reproduced verbatim from the pipeline output.

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*Generated by the `base_solver_ni_two_iterations` pipeline (Codex CLI, non-interactive; reasoning effort `ultra`; stages A.6 solve → A.8 verify → A.10 repair write-up; run `run_20260824_024352_d4c7ea4d, ec-2it-ultra`); two-iteration run, outcome from iteration 2. Independent verifier assessment: REJECTED with progress score 2/3 (major partial progress). Maintained by the AI in Mathematics & Operations Research project.

Setup and notation

Let $K^{-1} \le a_{ij}, b_{ji} \le K$, let $c_{ij} = a_{ij} b_{ji}$, and assume that every row and column of $C = (c_{ij})$ sums to n. Let S denote the set of stable matchings. For a stable matching μ, write

$$A_\mu = \sum_i X_{i\mu(i)}, \qquad B_\mu = \sum_j Y_{j,\mu^{-1}(j)}.$$

2 Fixed-matching conditional estimate

Fix a deterministic μ, relabel it as the identity, and condition on its matched Y-values $Y_{jj} = y_j$ and on μ being stable. Put

$$q_{ji} = 1 - e^{-b_{ji} y_j}, \qquad H = 1 + \sum_j (y_j \wedge 1).$$

The matched $X_i = X_{ii}$ are conditionally independent, with unnormalized densities

$$\phi_i(x) = a_{ii} e^{-a_{ii} x} \prod_{j \ne i} \big(1 - q_{ji}(1 - e^{-a_{ij} x})\big).$$

Since $q_{ji} \asymp_K y_j \wedge 1$,

$$Z_i \coloneqq \int_0^\infty \phi_i(x)\, dx \asymp_K H^{-1},$$

and

$$\Pr(X_i > x \mid y,\ \mu \text{ stable}) \le \begin{cases} C_K e^{-c_K H x}, & x \le 1, \\ C_K e^{-c_K (H + x)}, & x \ge 1. \end{cases} \tag{1}$$

Moreover, for a sufficiently small $c_0(K) > 0$,

$$\Pr(X_i \ge c_0/H \mid y,\ \mu \text{ stable}) \ge \frac{1}{2}.$$

Consequently,

$$\Pr\left(A_\mu < \frac{c_0 n}{4H} \,\middle|\, y,\ \mu \text{ stable}\right) \le e^{-c_K n}. \tag{2}$$

Let $T_k(z)$ denote the sum of the k largest coordinates of z. Applying the subset union bound to $W_i = \sqrt{H X_i}$, whose tails are uniformly exponential by (1), gives

$$\Pr\left(H T_k(X) > C_K \left[k \log^2 \frac{en}{k} + (t + \log k)^2\right] \,\middle|\, y,\ \mu \text{ stable}\right) \le e^{-t}. \tag{3}$$

3 The capped Bennett estimate

Take $M = D \log n$. For fixed $\delta, \eta > 0$ and $k = \lfloor \delta n \rfloor$, one obtains

$$\Pr\left(\frac{T_k(X \wedge M)}{A_\mu} > C_K \delta \log^2 \frac{e}{\delta} + \eta \,\middle|\, y,\ \mu \text{ stable}\right) \le C_{\delta,\eta,K}\, e^{-c_{\delta,\eta,K} n/M}. \tag{4}$$

The missing concentration details are as follows. Decompose

$$X_i \wedge M = (X_i \wedge 1) + V_i, \qquad V_i = \big((X_i \wedge M) - 1\big)_+.$$

The variables $H(X_i \wedge 1)$ have uniform exponential tails, so

$$HT_k(X \wedge 1) \le C_K \delta n \log \frac{e}{\delta} + \eta n$$

outside an event of probability at most $e^{-c_{\delta,\eta,K} n}$.

From (1),

$$\Pr(V_i > s \mid y,\ \mu \text{ stable}) \le p_H e^{-c_K s}, \qquad p_H \le C_K H e^{-c_K H}. \tag{5}$$

Choose

$$q_H = \left(c_K^{-1} \log \frac{p_H}{\delta}\right)_+, \qquad R_i = (V_i - q_H)_+.$$

Then

$$H\delta q_H + H\,\mathbb{E}R_i \le C_K \delta \log^2 \frac{e}{\delta}. \tag{6}$$

Furthermore,

$$\Pr(R_i > s) \le r_H e^{-c_K s}, \qquad r_H \le C_K \min\{\delta,\ He^{-c_K H}\},$$

and hence

$$\sum_i \mathrm{Var}(R_i) \le C_K n r_H. \tag{7}$$

For independent centered variables bounded above by M, Bennett's inequality states that

$$\Pr\Big(\sum_i Z_i > t\Big) \le \exp\left[-\frac{v}{M^2}\, h\Big(\frac{Mt}{v}\Big)\right], \qquad h(u) = (1+u)\log(1+u) - u, \tag{8}$$

where $v = \sum_i \mathrm{Var}(Z_i)$. Apply this with $Z_i = R_i - \mathbb{E}R_i$ and $t = \eta n / H$.

If H is bounded, the usual Bernstein consequence of (8) gives the exponent $c_{\delta,\eta,K} n/M$. If H is large, then by (7)

$$\frac{Mt}{v} \ge \frac{c_{\eta,K} M}{H^2 e^{-c_K H}},$$

so

$$\log\left(1 + \frac{Mt}{v}\right) \ge c_K H.$$

Since $h(u) \ge \frac{1}{2} u \log(1+u)$ for large u, (8) again gives

$$\Pr\left(\sum_i R_i - \mathbb{E}\sum_i R_i > \frac{\eta n}{H}\right) \le e^{-c_{\delta,\eta,K} n/M}.$$

Together with (2) and (6), this proves (4). Thus the Bennett issue is fully repaired.

Let

$$G_n = \Big\{\max_{i,j} (X_{ij} \vee Y_{ji}) \le M\Big\}.$$

For $D > 3K$,

$$\Pr(G_n^c) \le 2n^2 e^{-M/K} = o(1). \tag{9}$$

4 Complete conditional theorem

Assume additionally that

$$\boxed{\log \mathbb{E}|S| = o\left(\frac{n}{\log n}\right).} \tag{C}$$

Integrating (4) against the unnormalized law of $(Y(\mu), \{\mu \text{ stable}\})$, and then summing over deterministic matchings μ, yields

$$\Pr\left(\exists\, \mu \in S : \frac{T_{\lfloor \delta n \rfloor}(X(\mu))}{A_\mu} > C_K \delta \log^2 \frac{e}{\delta} + \eta\right) \le \Pr(G_n^c) + C_{\delta,\eta,K}\, \mathbb{E}|S|\, e^{-c_{\delta,\eta,K} n/\log n} = o(1). \tag{10}$$

The same argument applies to Y.

Let $A_{\mu,\delta}$ and $B_{\mu,\delta}$ be the paired-deletion truncations from the supplied truncated theorem. Since at most $\lfloor \delta n \rfloor$ coordinates are deleted,

$$A_\mu - A_{\mu,\delta} \le T_{\lfloor \delta n \rfloor}(X(\mu)), \qquad B_\mu - B_{\mu,\delta} \le T_{\lfloor \delta n \rfloor}(Y(\mu)).$$

Hence, with

$$r_{\delta,\eta} = C_K \delta \log^2 \frac{e}{\delta} + \eta,$$

we have, uniformly over $\mu \in S$ and with probability tending to one,

$$(1 - r_{\delta,\eta}) A_\mu \le A_{\mu,\delta} \le A_\mu, \qquad (1 - r_{\delta,\eta}) B_\mu \le B_{\mu,\delta} \le B_\mu. \tag{11}$$

The supplied Proposition 7.3 gives, for every fixed $\delta > 0$,

$$\sup_{\mu \in S} \left| \frac{A_{\mu,\delta} B_{\mu,\delta}}{n} - 1 \right| \xrightarrow{p} 0. \tag{12}$$

Combining (11) and (12),

$$\frac{A_{\mu,\delta} B_{\mu,\delta}}{n} \le \frac{A_\mu B_\mu}{n} \le \frac{A_{\mu,\delta} B_{\mu,\delta}/n}{(1 - r_{\delta,\eta})^2}.$$

For a prescribed final error, first choose fixed η and δ sufficiently small, and only then let $n \to \infty$. Therefore, under (C),

$$\boxed{\sup_{\mu \in S} \left| \frac{A_\mu B_\mu}{n} - 1 \right| \xrightarrow{p} 0.} \tag{13}$$

5 Complete homogeneous case

When $a_{ij} = b_{ji} = 1$, put

$$u_i = 1 - e^{-x_i}, \qquad v_j = 1 - e^{-y_j}, \qquad U = \sum_i u_i, \qquad V = \sum_j v_j.$$

For a fixed matching,

$$p_n = \int_{[0,1]^{2n}} \prod_{i \ne j} (1 - u_i v_j)\, du\, dv, \qquad \mathbb{E}|S| = n!\, p_n.$$

Since

$$\prod_{i\neq j}(1-u_iv_j)\le e^{-UV+\sum_i u_iv_i},$$

permutation invariance of the cube measure allows symmetrization under the integral:

$$\int e^{-UV+\sum_i u_iv_i}=\int e^{-UV}\,\frac{1}{n!}\sum_\pi e^{\sum_i u_iv_{\pi(i)}}.$$

Using $e^z\le 1+(e-1)z$ for $0\le z\le 1$, followed by Maclaurin's inequality,

$$\frac{1}{n!}\sum_\pi e^{\sum_i u_iv_{\pi(i)}}\le\sum_{k=0}^{n}(e-1)^k\frac{e_k(u)\,e_k(v)}{\binom{n}{k}}\le\exp\left(\frac{(e-1)UV}{n}\right).$$

Thus, for $n\ge 2$ and $\lambda_n=1-(e-1)/n>0$,

$$\mathbb{E}|S|\le n!\int_{[0,1]^{2n}}e^{-\lambda_n UV}\,du\,dv.$$

If f_n denotes the density of a sum of n independent uniforms, then

$$f_n(s)\le\frac{s^{n-1}}{(n-1)!}.$$

The exact substitution $p=UV$ gives

$$\mathbb{E}|S|\le\frac{n!}{((n-1)!)^2}\int_0^{n^2}p^{n-1}e^{-\lambda_n p}\log\frac{n^2}{p}\,dp.$$

The contribution from $p<1$ is $O(\log n/n)$, while for $p\ge 1$,

$$\int_1^{n^2}p^{n-1}e^{-\lambda_n p}\log\frac{n^2}{p}\,dp\le 2\log n\,\frac{\Gamma(n)}{\lambda_n^n}.$$

Since $\lambda_n^{-n}=O(1)$,

$$\boxed{\mathbb{E}|S|=O(n\log n).}\tag{14}$$

For $n=1$, $|S|=1$. Hence (C) holds, and (13) proves the complete homogeneous untruncated theorem.

6 Exact heterogeneous annealed reduction

A useful exact reduction, not present in the conditional count argument, is obtained by integrating one side first. For $y\in\mathbb{R}_+^n$, define

$$r_{ik}(y)=1-e^{-b_{ki}y_k},$$

and let Z_{ik} be independent Bernoulli variables with parameters $r_{ik}(y)$. Define

$$K_{ij}(y)=c_{ij}e^{-b_{ji}y_j}\,\mathbb{E}\left[\frac{1}{a_{ij}+\sum_{k\neq j}a_{ik}Z_{ik}}\right].\tag{15}$$

Direct row-by-row integration of the stability inequalities gives

$$\boxed{\mathbb{E}|S| = \int_{\mathbb{R}_+^n} \operatorname{per} K(y)\,dy.} \tag{16}$$

Indeed,

$$\prod_{k\neq j}\big(1 - r_{ik}(1-e^{-a_{ik}x})\big) = \mathbb{E}\exp\Big(-x\sum_{k\neq j} a_{ik}Z_{ik}\Big),$$

and integration against $e^{-a_{ij}x}$ produces (15).

This formula identifies the precise cancellation still needed. Set

$$A_i = \sum_k a_{ik}r_{ik}, \qquad V_i = \sum_k a_{ik}^2 r_{ik}(1-r_{ik}), \qquad d_{ij} = a_{ij}e^{-b_{ji}y_j}.$$

Uniformly as $A_i \to \infty$, a fourth-order expansion, with Bernstein control of the lower tail of the Bernoulli sum, gives

$$K_{ij}(y) = \frac{c_{ij}e^{-b_{ji}y_j}}{A_i}\left[1 + \frac{V_i}{A_i^2} - \frac{d_{ij}}{A_i} + O_K(A_i^{-2})\right]. \tag{17}$$

The two displayed corrections are both of order A_i^{-1}. On the important shell $A_i \asymp \log n$, retaining only one of them creates a multiplicative error of size

$$\exp\left(\Theta\left(\frac{n}{\log n}\right)\right),$$

which is exactly too large for (C). Bistochasticity suggests a cancellation after the permanent and the y-integration, but it does not give a pointwise cancellation. A uniform estimate

$$\log \int_{\mathbb{R}_+^n} \operatorname{per} K(y)\,dy = o\left(\frac{n}{\log n}\right) \tag{18}$$

has not been proved.

7 Why the fixed-matching race law cannot simply be optimized over μ

For a deterministic μ, exposing the complete Y-matrix gives

$$\Lambda_i^\mu = a_{i\mu(i)} + \sum_{\substack{j\neq\mu(i)\\ Y_{ji}<Y_{j,\mu^{-1}(j)}}} a_{ij}.$$

Then

$$\Pr_X(\mu \text{ stable} \mid Y) = \prod_i \frac{a_{i\mu(i)}}{\Lambda_i^\mu},$$

and, conditional on Y and on μ being stable, the variables

$$\Lambda_i^\mu X_{i\mu(i)}, \qquad i = 1,\dots,n,$$

are independent Exp(1) variables.

This statement fails for a data-dependent selected stable matching, and a 2×2 homogeneous example exhibits the selection bias explicitly. Suppose

$$w_1 : m_2 \succ m_1, \qquad w_2 : m_1 \succ m_2.$$

The swap matching σ is always stable, while the identity matching ι is stable exactly on $A \cap B$, where

$$A = \{X_{11} < X_{12}\}, \qquad B = \{X_{22} < X_{21}\}.$$

Thus the men-optimal selector is

$$\widehat{\mu} = \begin{cases} \iota, & \text{on } A \cap B, \\ \sigma, & \text{on } (A \cap B)^c. \end{cases}$$

Although the race rate for each man under the fixed matching σ is 1,

$$f_{X_{12}|\widehat{\mu}=\sigma}(x) = \frac{2}{3}\left(e^{-x} + e^{-2x}\right), \tag{19}$$

not e^{-x}. In fact,

$$f_{X_{12},X_{21}|\widehat{\mu}=\sigma}(x,z) = \frac{4}{3}e^{-x-z}\left(e^{-x} + e^{-z} - e^{-x-z}\right),$$

so the selected matched values are dependent. Therefore, the fixed-μ race identity does not supply the missing uniform inverse-rate estimate.

Conclusion

The following points are fully repaired:

- the fixed-matching conditional density and tail estimates;
- the capped Bennett step;
- the deduction from (C) and the truncated theorem;
- the complete homogeneous count and untruncated theorem;
- the exact fixed-matching race and annealed permanent identities.

The canonical heterogeneous conclusion

$$\sup_{\mu \in S} \left| \frac{A_\mu B_\mu}{n} - 1 \right| \xrightarrow{p} 0$$

is **not proved** from boundedness and bistochasticity alone. The exact remaining gap is either to prove (18), equivalently (C), or to establish a genuinely selection-stable uniform estimate for the inverse race rates. The 2×2 calculation above shows why the latter cannot be obtained merely by applying the fixed-matching conditional law to an extremal or worst stable matching.

References

[1] Itai Ashlagi, Mark Braverman, Geng Zhao. *Welfare Distribution in Two-sided Random Matching Markets*. Economics and Computation (EC), 2023. https://doi.org/10.1145/3580507.3597730